Top KMeans Models — Comparison Report

Input file: Unsupervised/outputs/kmeans_benchmark/unsupervised_experiments_results.csv

Rows (model variants): 18

Families represented in report: 4

Included families: kmeans_feature_subset, kmeans_k_sweep, kmeans_pca, kmeans_transformer_weight

Top family by silhouette score:

- family: kmeans_pca
- variant: pca=2
- silhouette: 0.3373

KMeans Family Descriptions (1/2)

1. kmeans_k_sweep

Tests k=2 to k=10 on all features (baseline KMeans tuning).

Purpose: find the best cluster count without changing the feature space.

Best model:

  Variant: k=2

  Silhouette: 0.0684

  NMI: 0.0001, ARI: -0.0001

2. kmeans_feature_subset

Tests clustering with focused feature domains:

  - all_features: All available features (post-exclusion)

  - psychosocial_focus: Sleep, Physical Activity, Diet, Social Support, Substance Use, Counseling Service Use, Family History, Chronic Illness, Financial Stress

  - academic_lifestyle_focus: Course, CGPA, Semester Credit Load, Extracurricular Involvement, Residence Type, Relationship Status, Age, Gender

Purpose: identify whether specific feature groups separate students better than full features.

Best model:

  Variant: psychosocial_focus

  Silhouette: 0.1049

  NMI: 0.0003, ARI: 0.0003

KMeans Family Descriptions (2/2)

3. kmeans_pca

PCA dimensionality reduction followed by KMeans with k=4.

Tests n_components: 2, 5, 10, 20 on all features.

Purpose: reduce high-dimensional one-hot feature space into compact latent components.

Interpretation: lower components simplify structure; higher components retain more detail.

Best model:

  Variant: pca=2

  Silhouette: 0.3373

  NMI: 0.0001, ARI: -0.0001

  CH: 6963.9953, DB: 0.8903

4. kmeans_transformer_weight

Tests weighted preprocessing to emphasize numeric vs categorical features.

  - num_heavy (1.5x numeric, 1.0x categorical)

  - cat_heavy (1.0x numeric, 1.5x categorical)

Purpose: test whether burnout patterns are driven more by numeric intensity vs category identity.

Interpretation: num_heavy boosts scaled numeric signals; cat_heavy boosts one-hot category influence.

Best model:

  Variant: num_heavy

  Silhouette: 0.0923

  NMI: 0.0003, ARI: -0.0002

  CH: 1006.3909, DB: 2.2806

Final Unsupervised Insights

Overall model performance summary:

- Clustering finds structure, but alignment to burnout quartiles is weak (ARI/NMI near zero across families).
- This means clusters are useful for segmentation, not strong burnout prediction as-is.
- PCA-based variants (especially low components) improve geometric separation the most.
- kmeans_k_sweep runs are highly stable across seeds for many k values.

Recommended interpretation:

- Use these models for student profile discovery and intervention design by segment.
- Do not use cluster labels alone as a clinical/operational burnout predictor.

Best geometric variant observed:

- family: kmeans_pca
- variant: pca=2
- silhouette: 0.3373
- davies_bouldin: 0.8903

KMeans Cluster Interpretation

How to interpret KMeans clusters (personas):

- We distinguish clusters by comparing per-cluster feature means vs overall means.
- Numeric means come from kmeans_cluster_profile_summary.csv (Age, CGPA, Financial_Stress, Credit_Load).
- We verify structure using silhouette/DB/CH and visualize overlap via PCA and heatmaps.

Cluster personas from profile means:

- Cluster 0: higher financial stress, higher credit load, lower age, average CGPA
- Cluster 1: lower financial stress, lower credit load, lower age, average CGPA
- Cluster 2: lower financial stress, higher credit load, lower age, average CGPA
- Cluster 3: average financial stress, lower credit load, higher age, average CGPA

Why these are clusters (and limits):

- silhouette=0.0579, davies_bouldin=3.0014, calinski_harabasz=590.2788
- ARI=-0.0002, NMI=0.0003 vs burnout quartiles.
- Conclusion: useful segmentation personas, but weak direct burnout-label alignment.

Categorical-feature findings:

- Categorical differences are summarized by top category and lift per cluster-feature pair.
- Cluster 3: Residence_Type=On-Campus (41.6% vs 40.0%, lift=1.04)
- Cluster 0: Gender=Male (52.5% vs 50.6%, lift=1.04)
- Cluster 3: Course=Medical (31.1% vs 30.0%, lift=1.04)
- Cluster 0: Residence_Type=Off-Campus (41.0% vs 39.7%, lift=1.03)

PCA Component Interpretation (PC1/PC2)

PC1/PC2 meaning in this project:

- Explained variance: PC1=0.0944, PC2=0.0920

Top weighted features in PC1:

- CGPA: 0.670
- Age: 0.539
- Financial_Stress: -0.509
- Semester_Credit_Load: 0.020

Top weighted features in PC2:

- Semester_Credit_Load: 0.873
- Age: 0.346
- Financial_Stress: 0.339
- CGPA: -0.046

Interpretation:

- PC1/PC2 are weighted combinations of original processed features, not raw columns.
- They summarize dominant variation for visualization and can highlight which variables drive separation.

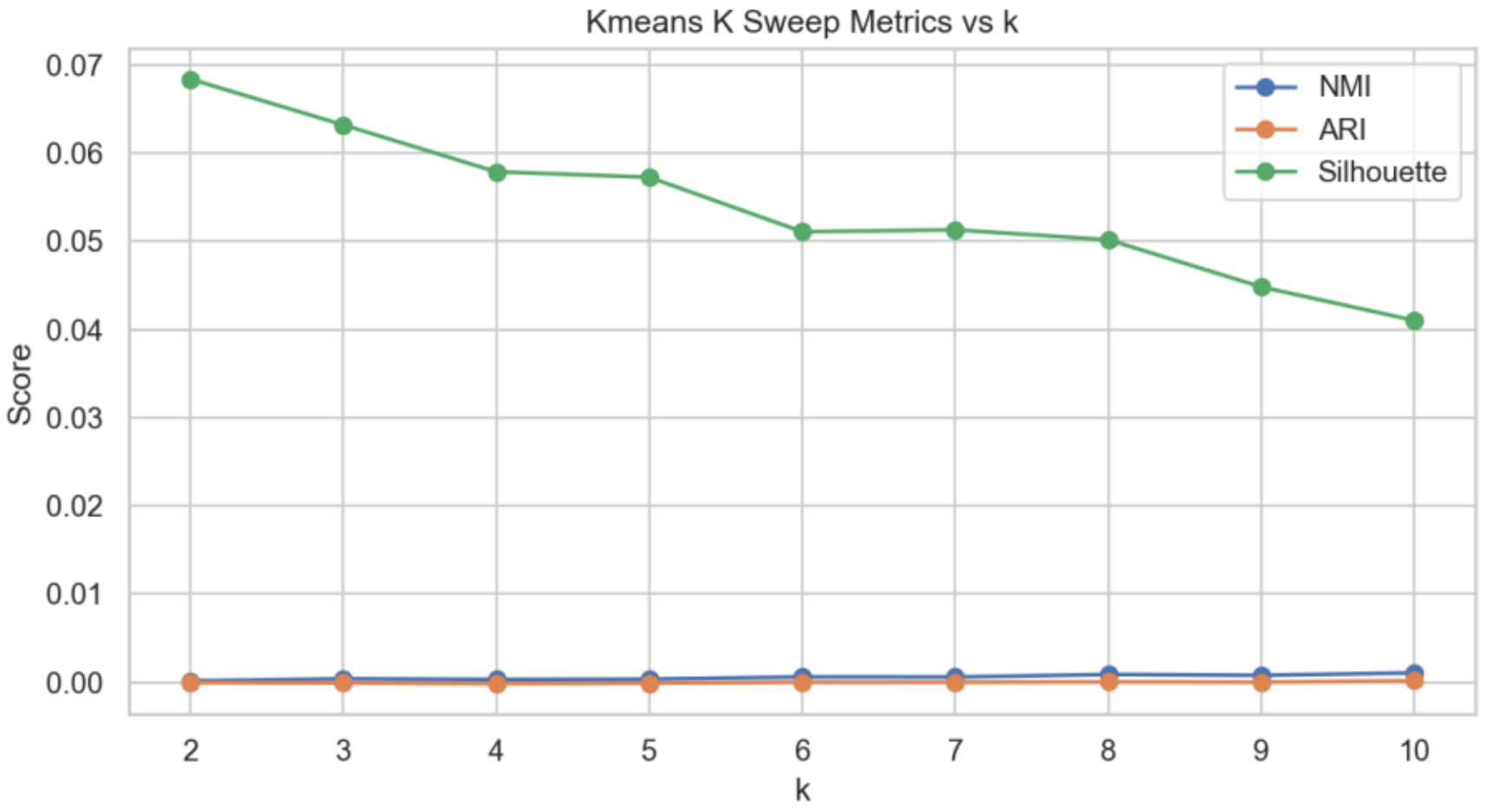
Top Models by Silhouette Score (4 Families)

Metric meanings:

- silhouette: cluster separation (higher better)
- davies_bouldin: cluster overlap (lower better)
- calinski_harabasz (CH): compactness vs separation (higher better)
- adjusted_rand_index (ARI): alignment to burnout quartiles (higher better)
- normalized_mutual_info (NMI): shared information with burnout quartiles (higher better)

rank	family	variant	k	silhouette	davies bouldin	calinski harabasz	adjusted rand index	normalized mutual info
1	kmeans_pca	pca=2	4	0.3373	0.8903	6963.9953	-0.0001	0.0001
2	kmeans_feature_subset	psychosocial_focus	4	0.1049	2.7097	1140.6617	0.0003	0.0003
3	kmeans_transformer_weight	num_heavy	4	0.0923	2.2806	1006.3909	-0.0002	0.0003
4	kmeans_k_sweep	k=2	2	0.0684	3.5979	762.2479	-0.0001	0.0001

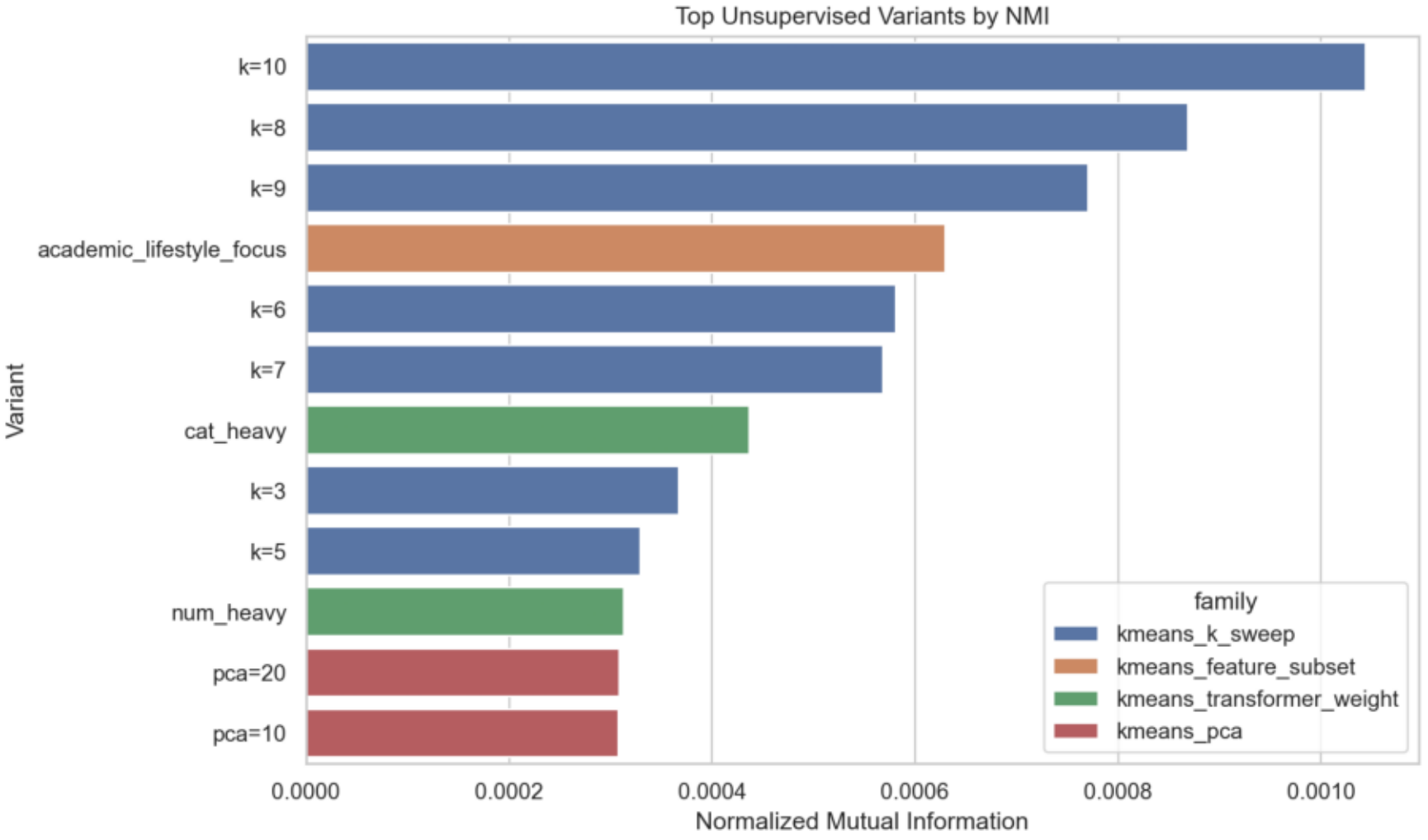
Experiments Figure: kmeans_k_sweep_metrics.png



What this figure shows:

Line chart across k=2..10 showing NMI, ARI, and Silhouette for the KMeans k-sweep family.

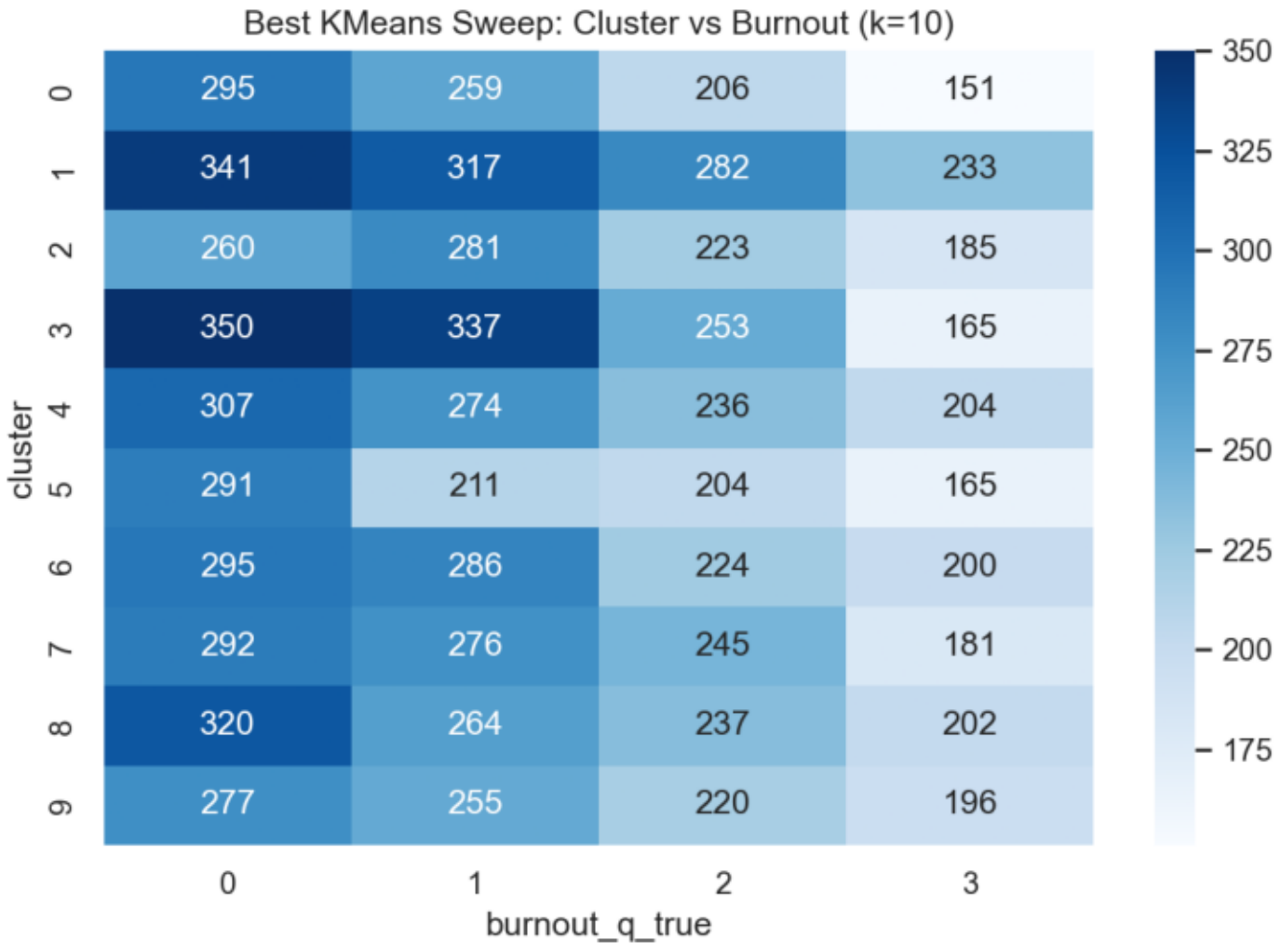
Experiments Figure: top_variants_by_nmi.png



What this figure shows:

Bar chart of top model variants ranked by NMI; compares alignment with burnout quartiles.

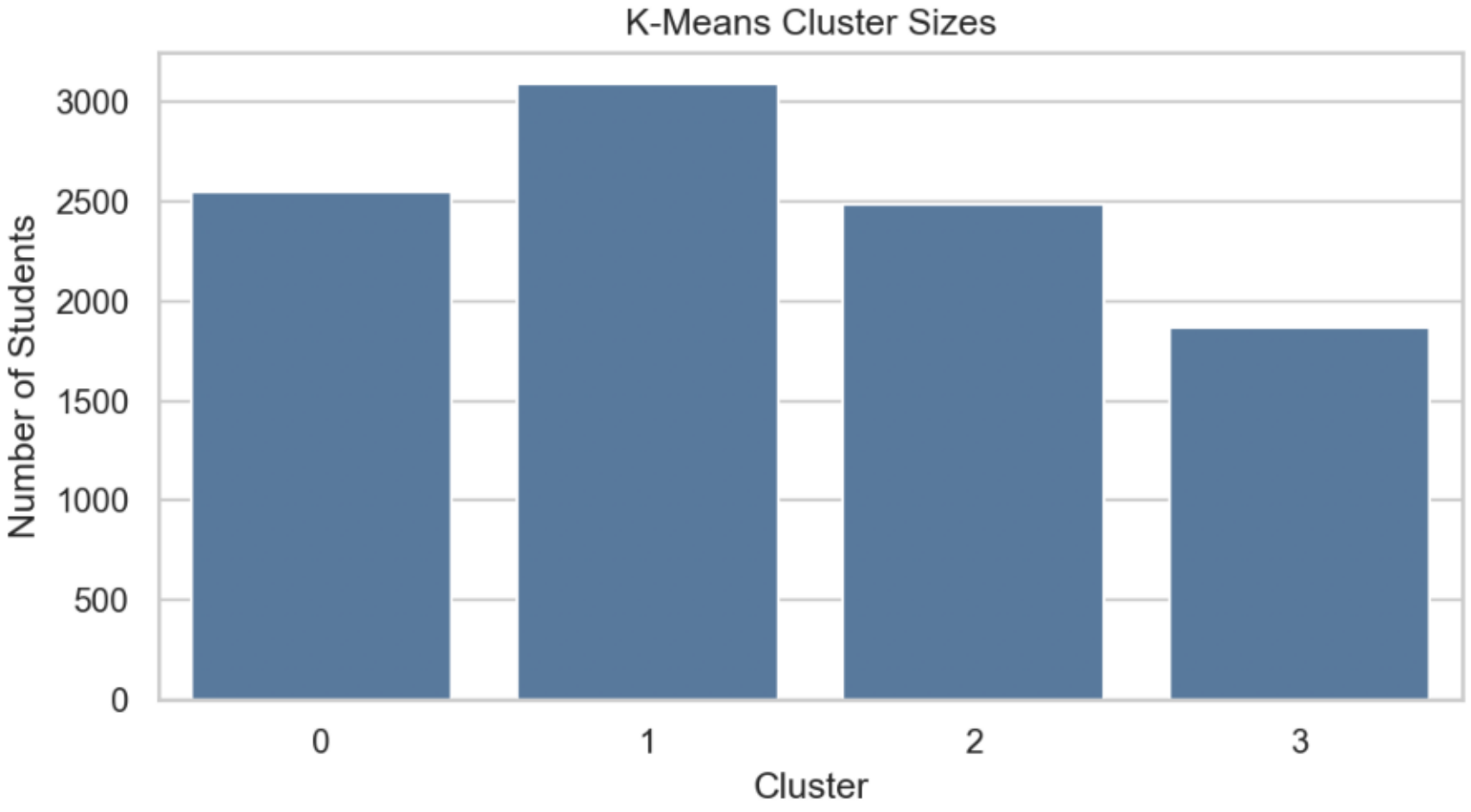
Experiments Figure: best_kmeans_cluster_vs_burnout.png



What this figure shows:

Heatmap of discovered clusters vs true burnout quartiles for the best kmeans_k_sweep run.

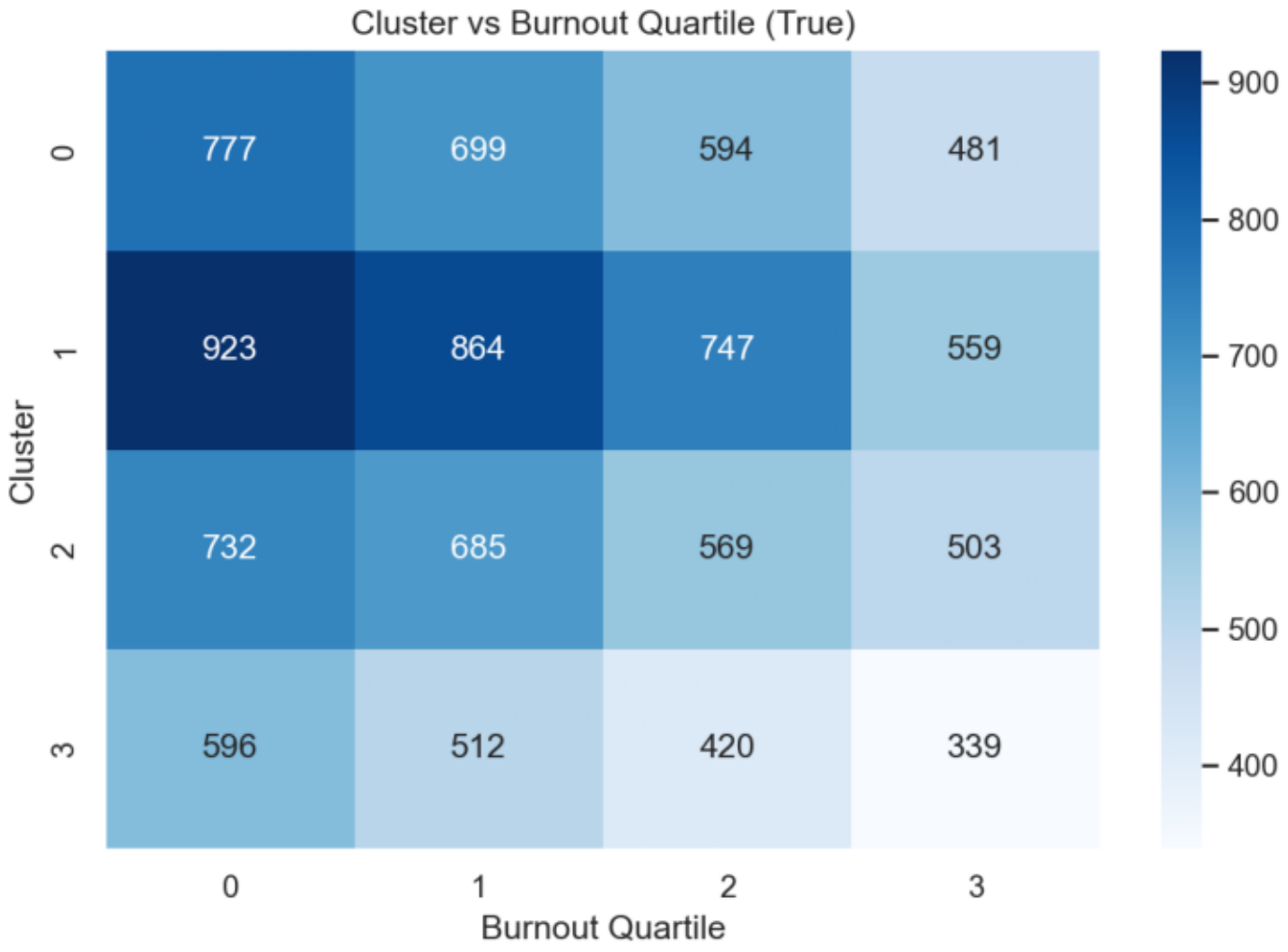
KMeans Figure: cluster_sizes.png



What this figure shows:

Bar chart of number of students per cluster; useful to spot highly imbalanced clusters.

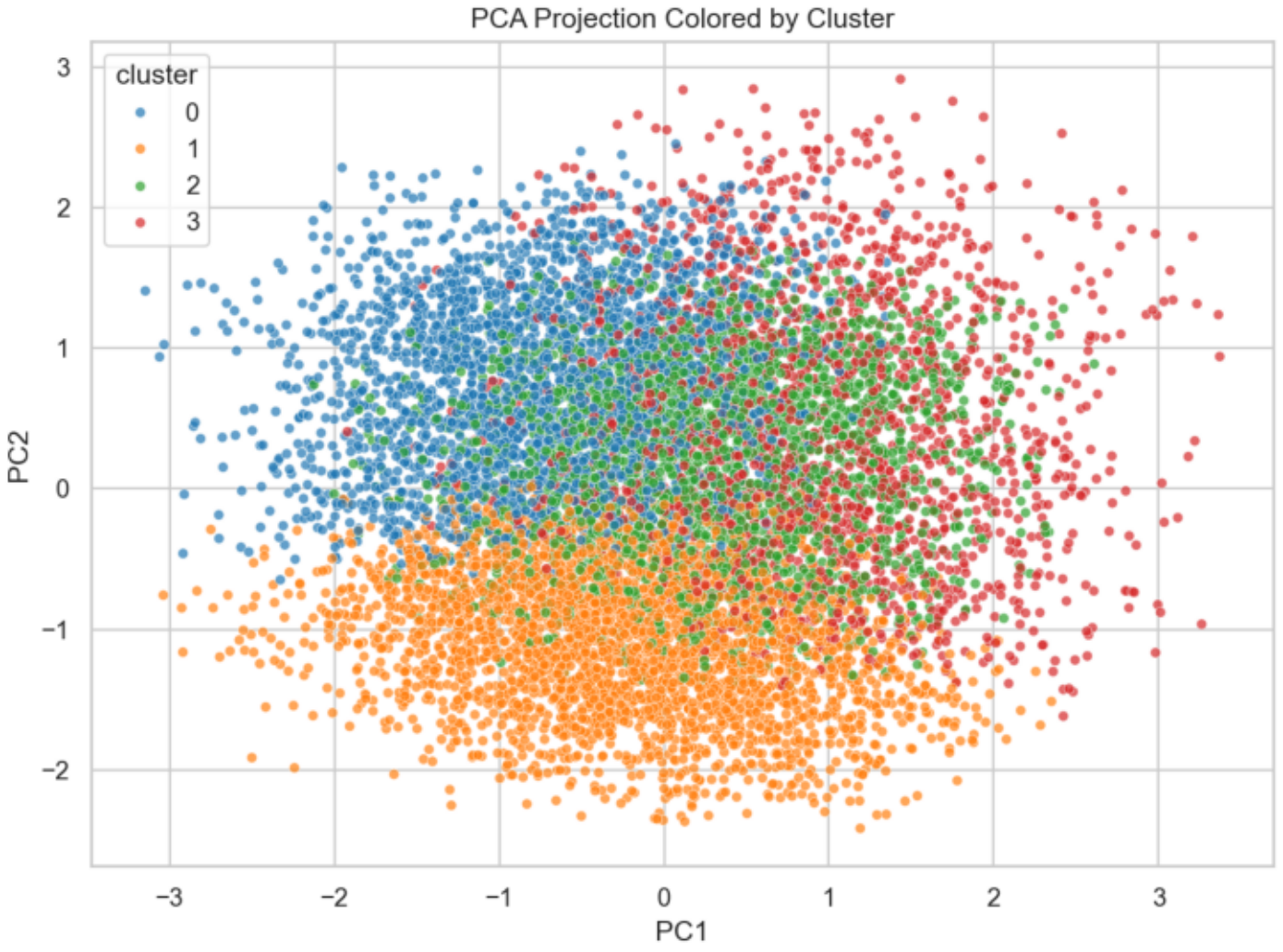
KMeans Figure: cluster_vs_burnout_heatmap.png



What this figure shows:

Heatmap crossing discovered clusters with true burnout quartiles; highlights where clusters align or mix quartiles.

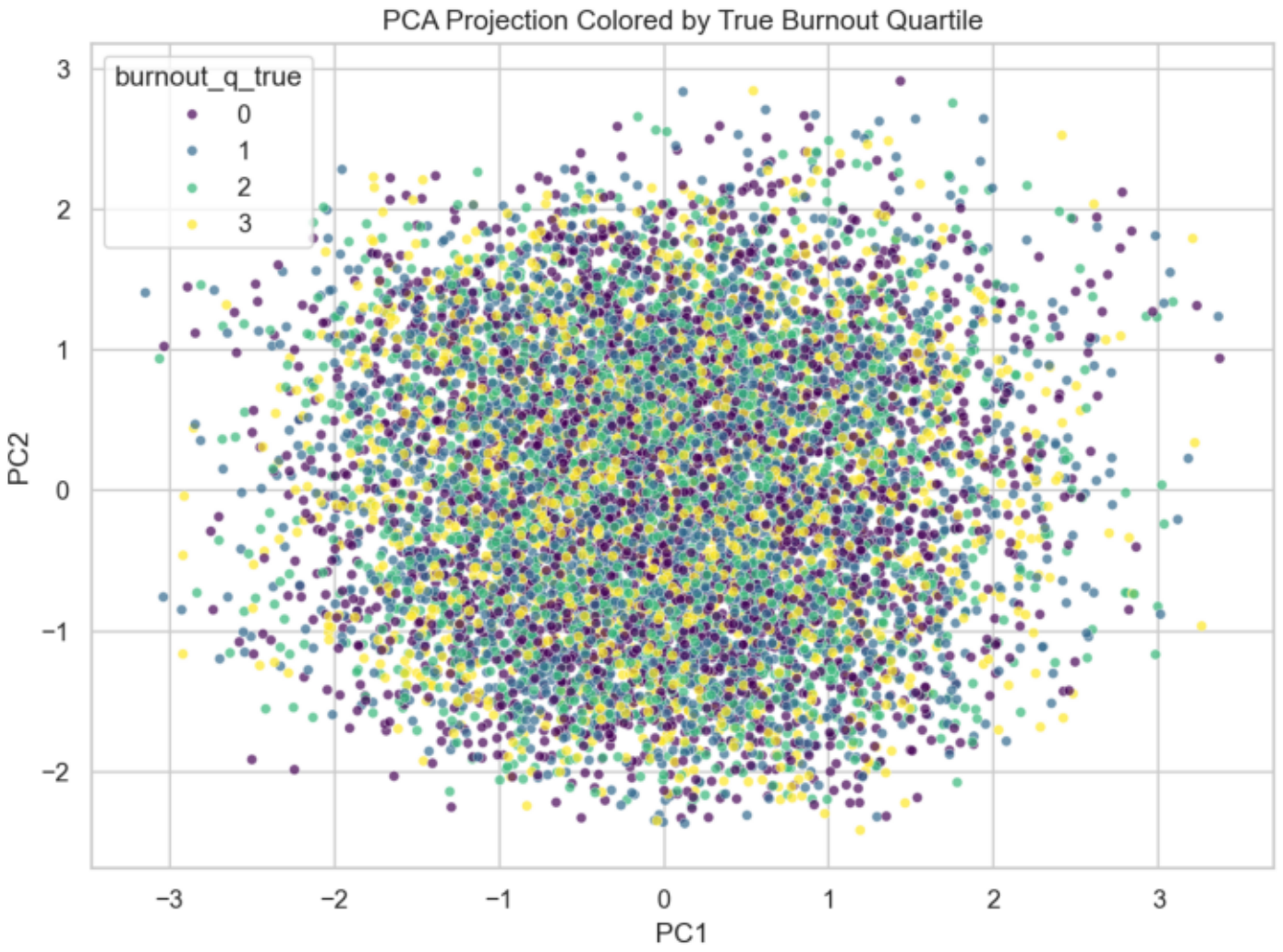
KMeans Figure: pca_clusters.png



What this figure shows:

2D PCA projection colored by cluster label; gives a visual sense of separation/overlap in reduced space.

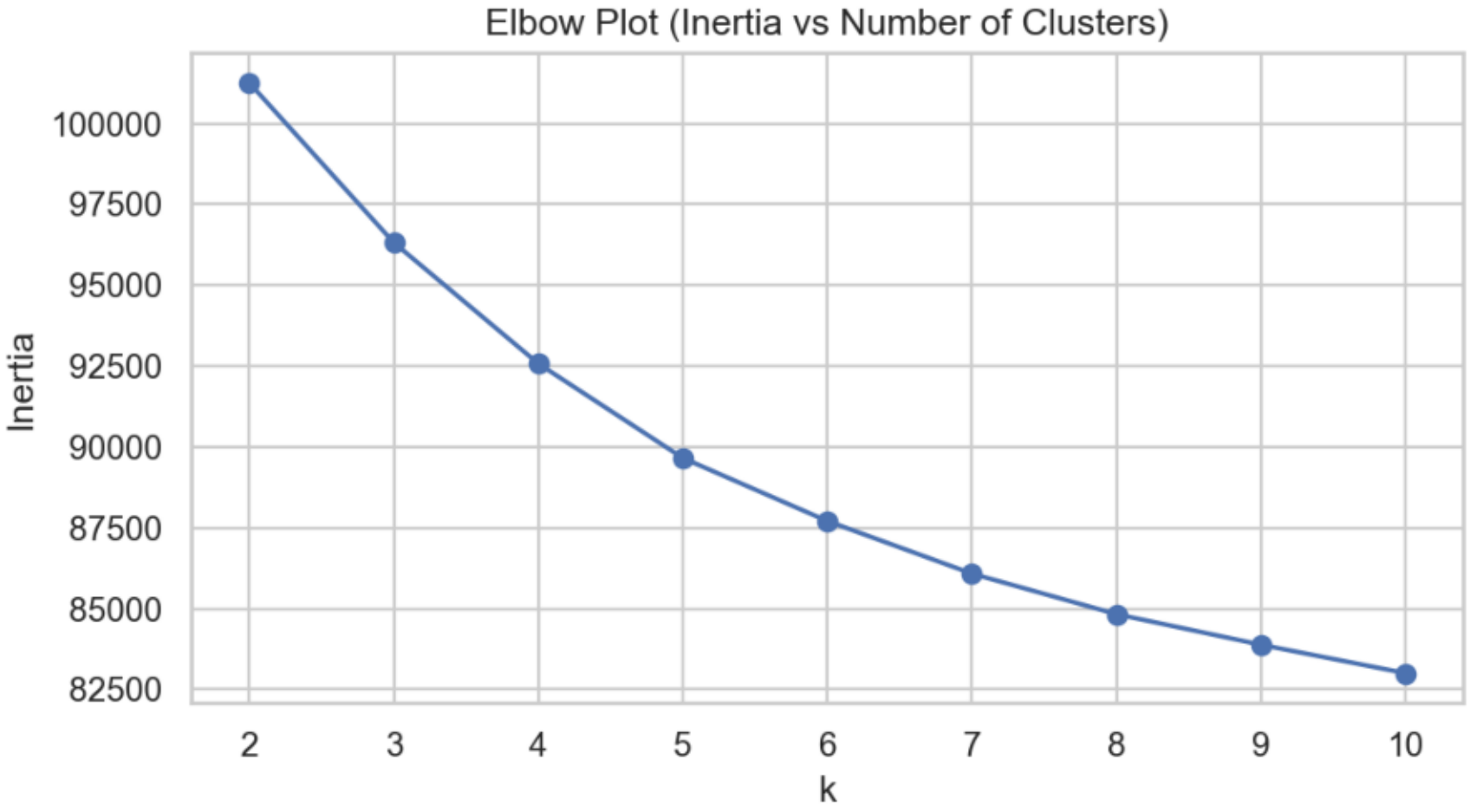
KMeans Figure: pca_true_burnout.png



What this figure shows:

Same PCA projection colored by true burnout quartiles; compare against cluster-colored view to assess alignment.

KMeans Figure: elbow_plot.png



What this figure shows:

Inertia vs k curve for k=2..10; look for an elbow where adding more clusters yields diminishing returns.